

GROUP 7

Peyala Café

How much to prepare, and from whom to buy

Demand sensing, newsvendor prep, safety stock, sourcing and a reorder trigger, on a six-item extract of 24,031 order lines

An alumnus of our own programme opened the engagement and shared the data

₹39.4 L

annual revenue; ₹75,857 a week

62.1%

gross margin on the extract

44.4%

of revenue is pizza alone; top two items 59.1%

12,000

bills, 24,031 lines, Jan 2023 – Dec 2024

HOW WE GOT IN

Priyesh Mandal, PGP 2024–26, IIM Visakhapatnam; now Domain Consultant, Manufacturing at Infosys. He ran a prior engagement with Peyala and shared POS, inventory and supplier data, with no defined problem.

THE CLIENT

Peyala Café, Daak Bunglow Road, Medinipur. "Café & Adda", trading 2:00 pm – 11:30 pm, own Android ordering app, Zomato award holder. Our extract covers six perishable items, not the full menu.

PROBLEM DEFINITION

Peyala prepares six perishable menu items to a daily judgement-based prep plan and buys ingredients from three suppliers selected on headline delivery price alone. The two decisions are made by different people, on different cadences and are never reconciled.

How much should Peyala prepare each day, when and from whom should it replenish, and what does the disconnect between the kitchen and the purchase desk cost in rupees?

ALREADY COVERED

- Comparative forecasting and breakeven (covered by the 2025 engagement).
- Peak-hour staffing: the timestamp is synthetic.
- Pricing : no volume response to discount.

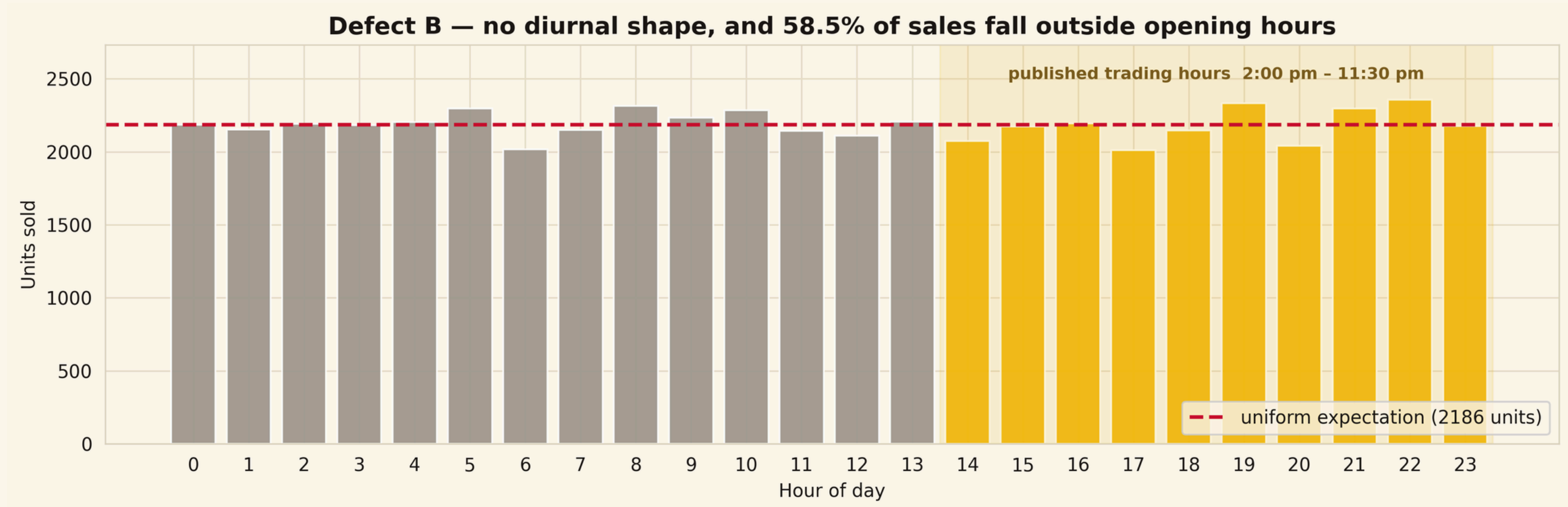
FOUR SUB-QUESTIONS

- Q1. demand sensing (ARIMAX)
- Q2. prep quantity (newsvendor)
- Q3. replenishment and sourcing (safety stock, LP, TOPSIS)
- Q4. automation (RPA reorder trigger).

Three files and four defects that bound what they can support

WHAT WE RECEIVED	DEFECT	HOW WE HANDLED IT
POS transactions Order line, Jan 2023 – Dec 2024. Demand levels, revenue mix, baskets.	A 28% of POS lines fail $\text{Total} = \text{Qty} \times \text{Price} - \text{Discount}$	Treat <code>Total_Amount</code> as authoritative
Inventory & prep log Item $\times$ day, 366 days. Prep-versus-actual behaviour and spoilage only.	B Hourly timestamp is synthetic, no diurnal shape at all	No peak-hour or staffing work; Date retained
Supplier master Supplier $\times$ material. Price, lead time, MOQ, capacity, LP and TOPSIS inputs.	C Inventory log flattens every item to ~29–30 units/day	POS for demand levels; log for behaviour only
	D Delivery_Cost_INR is per delivery; 14 duplicate Route_IDs	$\div$ MOQ $\rightarrow$ ₹6.75–₹21.36/unit; deduplicate

58.5% of units are logged while the café is shut, the timestamp is synthetic



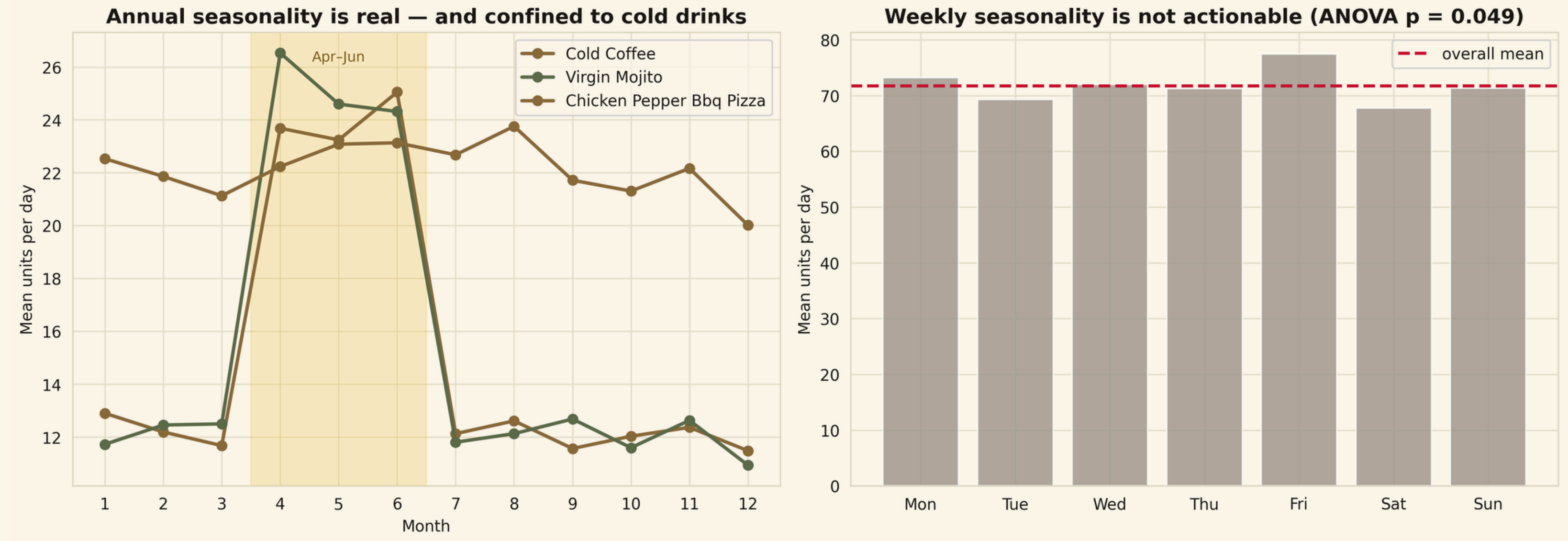
OBJECTIVE

Test whether the POS timestamp can support any time-of-day analysis, by comparing hourly unit volumes against the café's published 2:00 pm – 11:30 pm trading hours.

INTERPRETATION

Volume is near-uniform across all 24 hours (CV 4.4%), and 7,051 of 12,000 bills fall outside opening hours. The field is generated, not observed, so peak-hour, staffing and queueing analysis is inadmissible.

Two drinks nearly double in April - June, no weekday cycle is actionable



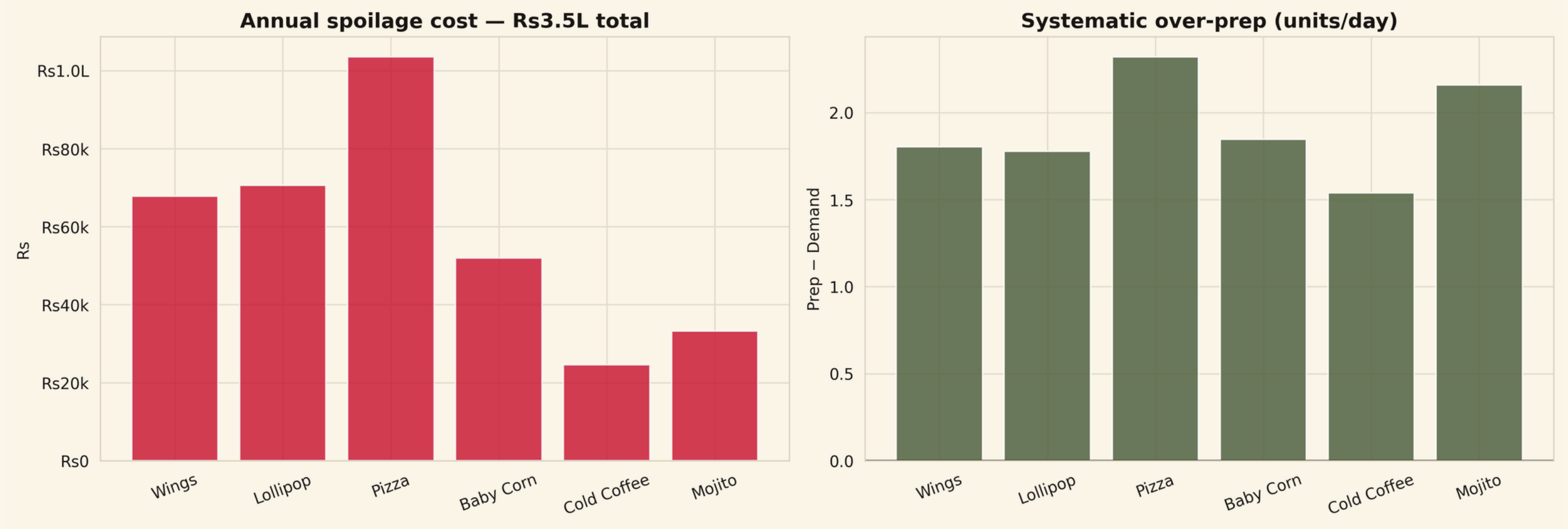
OBJECTIVE

Identify which seasonal cycles are strong enough to model: an annual (summer) effect, a weekday effect, or both.

INTERPRETATION

Cold Coffee +98% and Virgin Mojito +109% in Apr–Jun; the weekday spread is ~14% at $p = 0.049$. So we carry an annual Summer regressor and fit no SARIMA with $s = 7$.

Spoilage costs ₹3.5 lakh a year, and the café still runs short on 27% of days



OBJECTIVE

Put a rupee value on what the current prep plan wastes, and measure whether the kitchen over- or under-prepares by item.

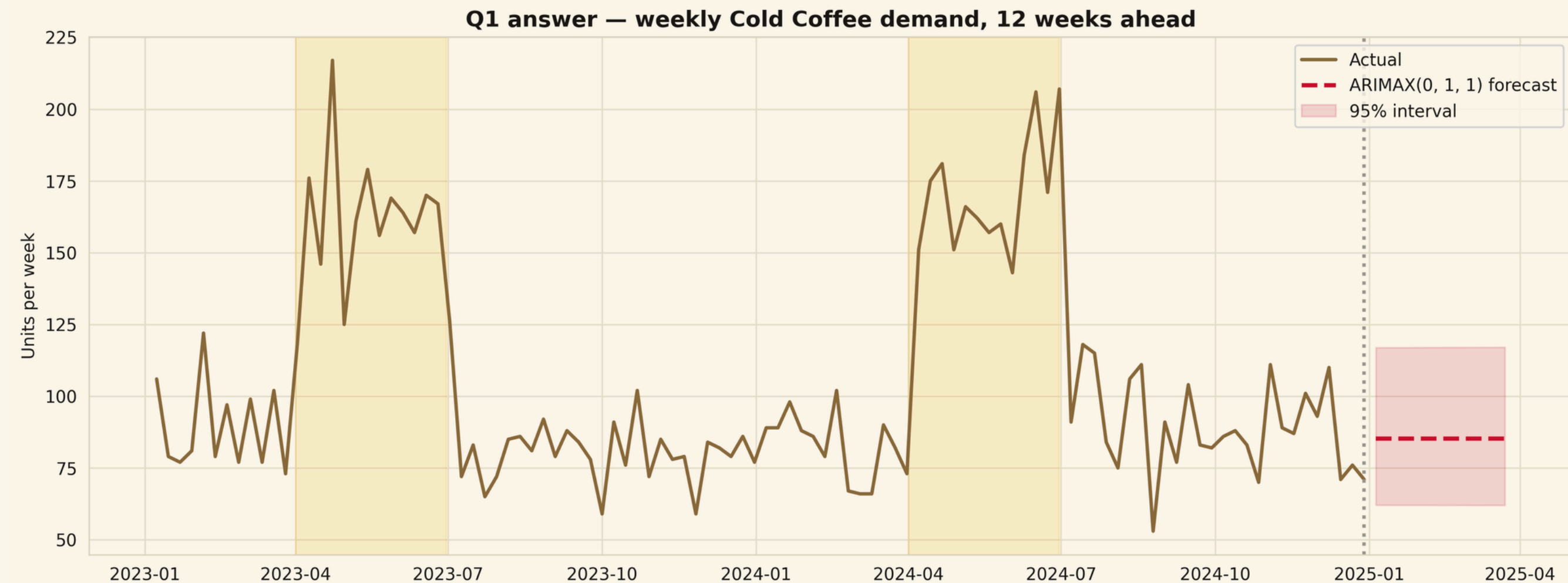
INTERPRETATION

9% of everything prepped spoils (₹3.5 lakh/yr) while shortages persist on 27% of days, with a consistent over-prep bias of +1.5 to +2.3 units, a judgement plan, not a costed one.

Four sub-questions, four techniques, four course anchors

	DECISION	TECHNIQUE	COURSE ANCHOR
Q1	What will next week's demand be?	ARIMAX(0,1,1) on log demand + Summer flag; rolling-origin validation	Time-series forecasting
Q2	How much should the kitchen prepare?	Newsvendor, item-specific critical ratios, scored on 366 days	Inventory under uncertainty
Q3	When and from whom to replenish?	Safety stock and ROP · LP allocation · TOPSIS sensitivity	Inventory · optimisation · MCDM
Q4	Who watches stock every day?	Rule-based reorder trigger on corrected reorder points	Process automation, Sessions 07–08

Demand is seasonal, not weekly: a summer flag cuts forecast error by 51%



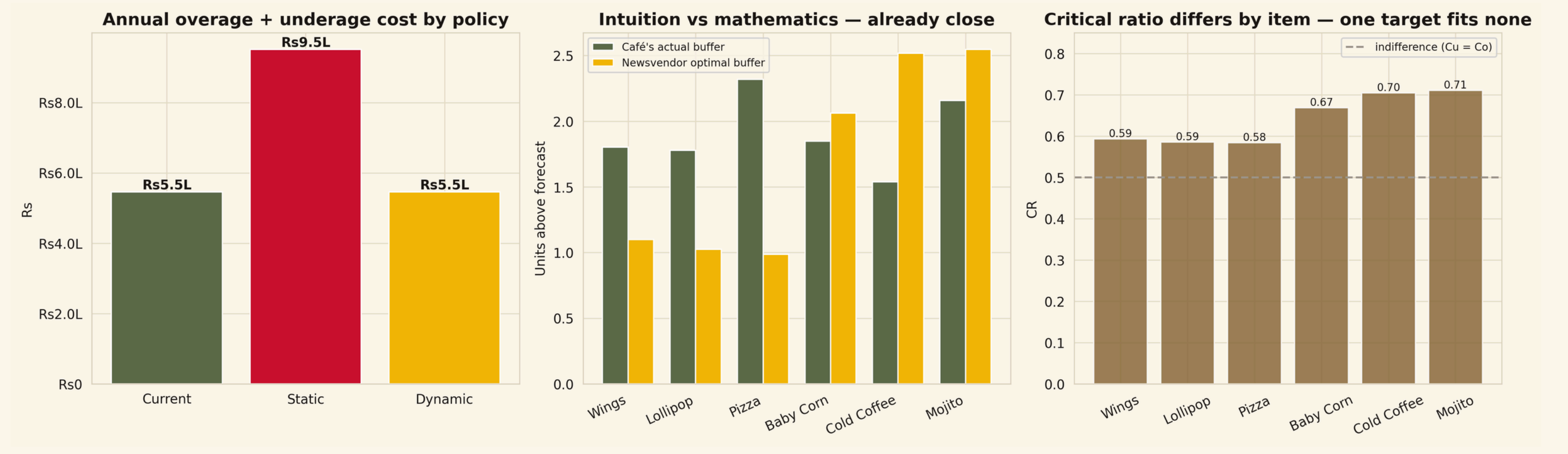
OBJECTIVE

Produce a weekly demand forecast good enough to drive a prep decision, and prove the gain is real rather than an artefact of one lucky test window.

INTERPRETATION

ARIMAX(0,1,1)+Summer: coefficient 0.673 $\Rightarrow \times 1.96$, $p < 0.001$; AIC 691.4 vs 761.0. Rolling-origin RMSE 16.9 vs 34.7 (5 folds $\times$ 8 weeks); the 80/20 split shows only 6% because Aug–Dec contains no summer.

A fixed daily prep quantity would cost ₹4.0 lakh a year more than the café's own judgement



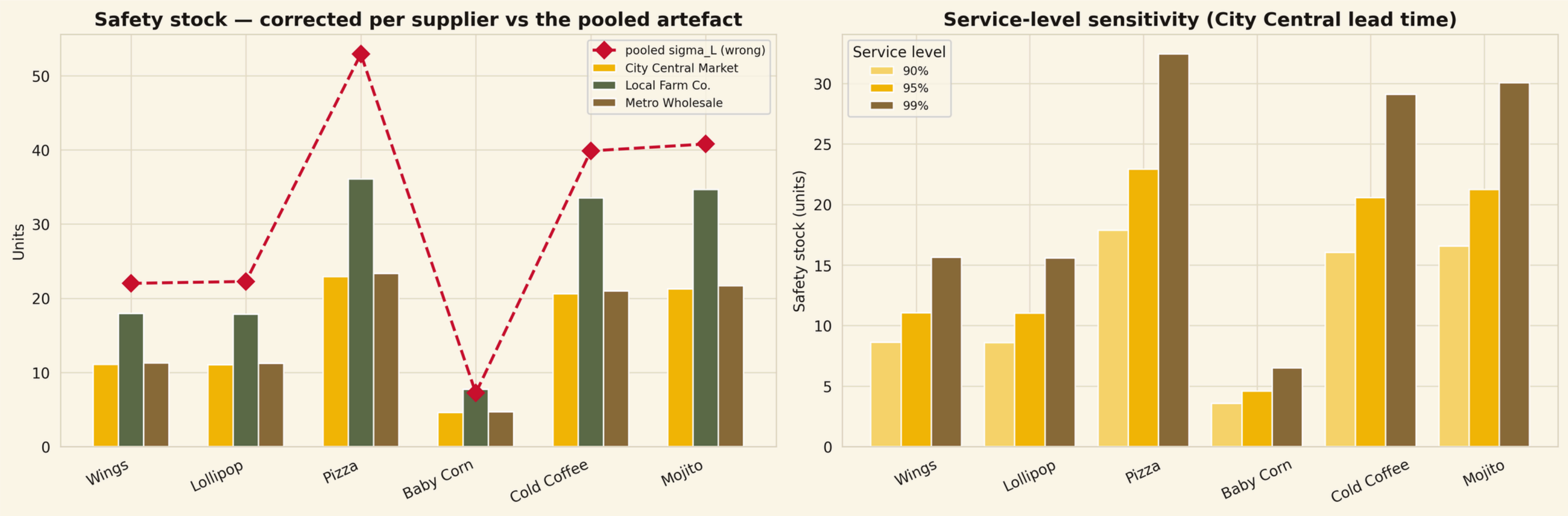
OBJECTIVE

Score three prep policies, current judgement, a static newsvendor Q^* , and a dynamic newsvendor on forecast error, on identical days and one cost function.

INTERPRETATION

Current ₹5.5 L · Static ₹9.5 L · Dynamic ₹5.5 L. The fixed quantity is +74% worse because the critical ratio differs by item (0.58–0.71) and the café's buffer is already within ~1 unit of optimal. This is a cost avoided, not a saving found.

Corrected safety stock ranges from 5 to 36 units, not 69 for everything



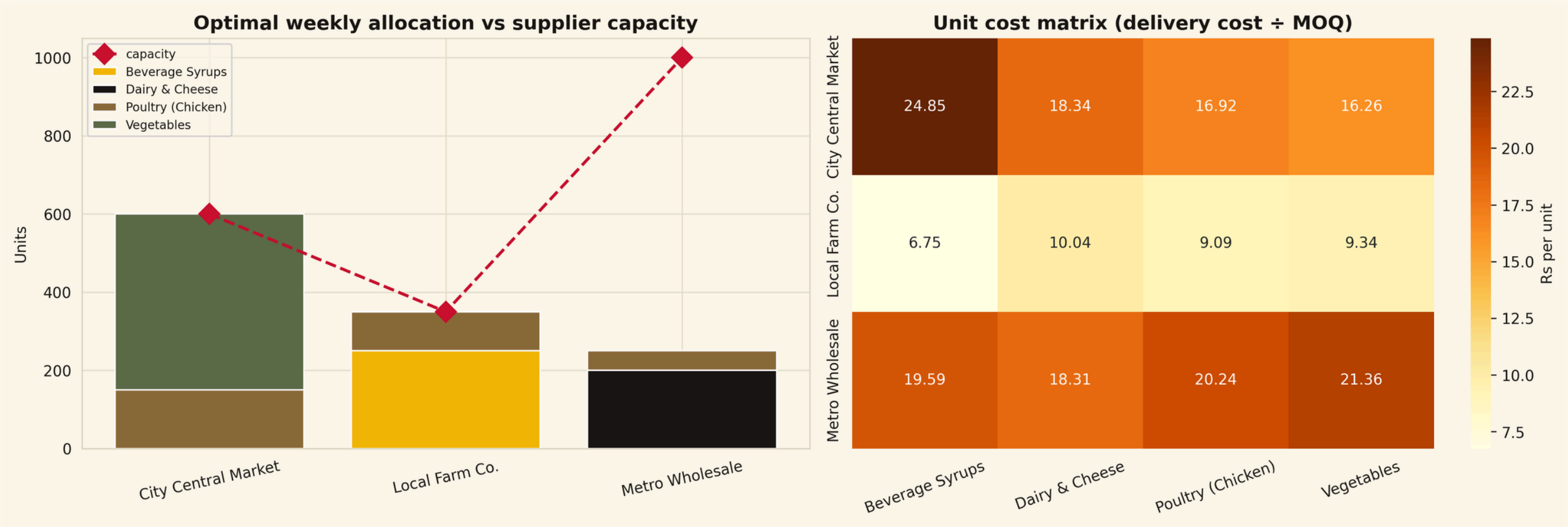
OBJECTIVE

Set safety stock and a reorder point per item at a 95% service level, and test whether a pooled lead-time variance is admissible.

INTERPRETATION

Pooled $\sigma(\text{lead time})$ is 92% between-supplier spread, so it returns ~69 units for every item alike. Decomposed, cover is item-specific: 5 units for Baby Corn to 36 for Pizza, the pooled figure over-provisions the small movers by ~13 $\times$.

Capacity, not lead time, is the binding constraint: by two independent routes



OBJECTIVE

Allocate weekly purchases across three suppliers at least cost under capacity limits, and price the cost-versus-lead-time trade-off in rupees rather than asserting it.

INTERPRETATION

The lead-time premium costs ₹774/yr in holding; the price advantage is worth ₹1.8 lakh/yr , a 227× ratio. The LP optimum is ₹17,127/week (22.6% of revenue) with Local Farm at its ceiling and a shadow price of ₹11.15. Both routes say: negotiate capacity. TOPSIS rankings flip with weighting, so no single winner is claimed.

Four decisions, four defensible answers, and what they cannot be used for

Peyala Café — Operations Analytics: results at a glance

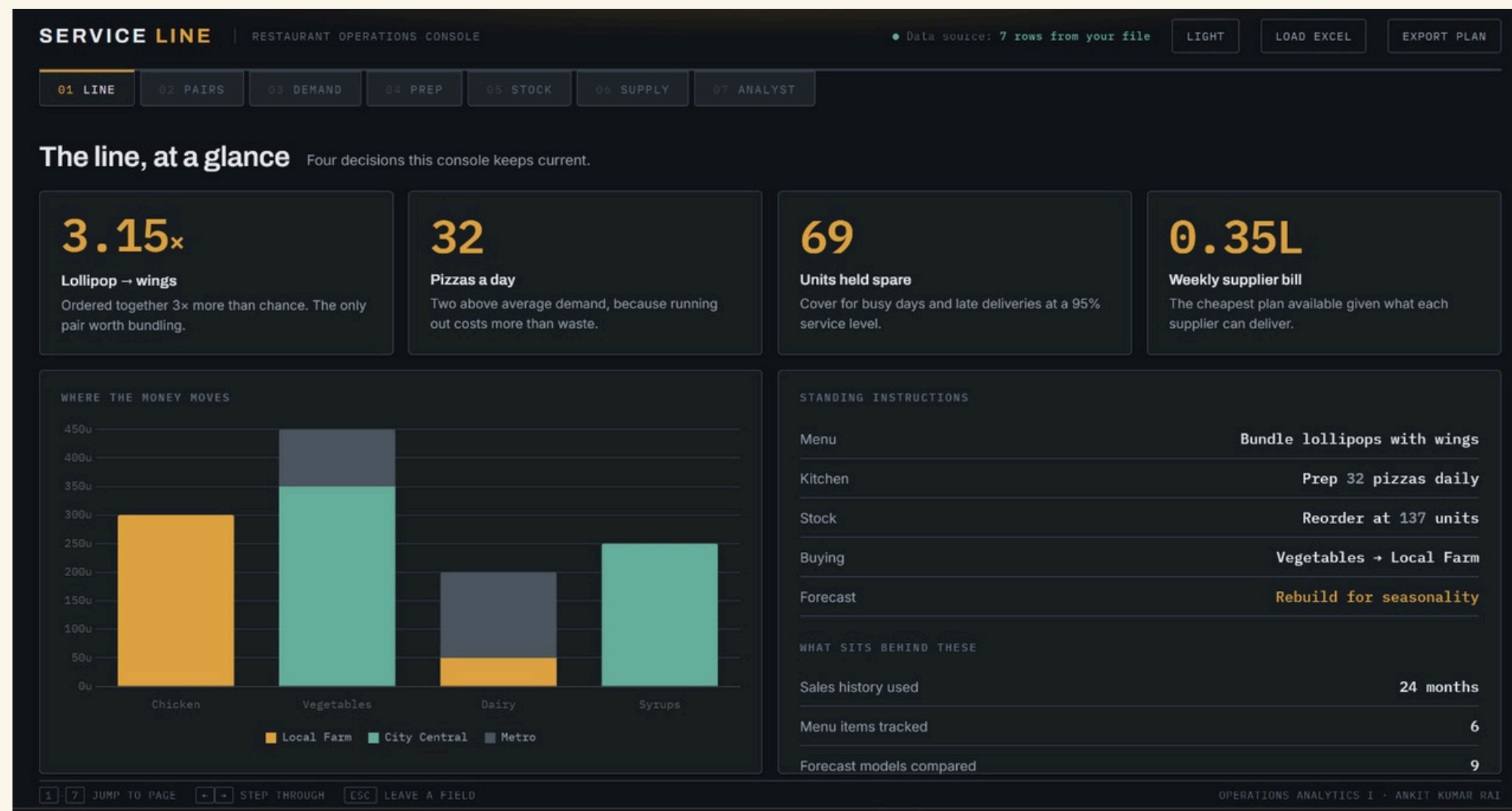


- Q1** Forecast error 51% lower under rolling-origin validation
- Q2** ₹4.0 lakh/yr avoided by not adopting a fixed prep quantity
- Q3** Safety stock 5–36 units; ₹17,127/week LP optimum, capacity-bound
- Q4** Reorder trigger specified on the corrected reorder points

LIMITATIONS

- Six items, not the full menu
- no time-of-day inference (Defect B)
- inventory levels unusable
- TOPSIS is weight-sensitive
- the ×1.96 summer uplift rests on two annual cycles.

Turns today's sales into tomorrow's prep, reorder and purchase decisions, every day



Seven tabs : Line · Pairs · Demand · Prep · Stock · Supply · Analyst. Loads the café's Excel export, recomputes every figure, exports a plan. The models are the ones in this deck.

Roadmap: live POS feed · recipe BOM · scheduled alerts (Session 07–08 RPA pattern) · Bengali interface for kitchen staff.

RECOMMENDATIONS

Five actions, each with an owner and its evidence

ACTION	OWNER	HORIZON	EVIDENCE
Keep judgement-based daily prep; do not standardise to a fixed quantity	Kitchen lead	Immediate	Q2 · ₹4.0 L avoided
Adopt item-specific safety stock and reorder points	Purchase desk	2 weeks	Q3a · 5–36 units
Negotiate additional weekly capacity with Local Farm	Owner	1 month	Q3b · shadow price −₹11.15
Bundle Lollipop with Wings; stop discounting items with no volume response	Owner	1 month	Lift 3.15 · all p > 0.5
Repair the POS timestamp and the delivery-cost field at source	App vendor	1 quarter	Defects B and D



Thank you